



Fluorescence imaging using a fluorescent protein with a large Stokes shift

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ABSTRACT

Keima is a far-red fluorescent protein endowed with a large Stokes shift. It absorbs light maximally at around 440 nm and emits maximally at around 620 nm. While the original Keima is obligately tetrameric (tKeima), the dimeric and monomeric versions (mKeima and dKeima, respectively) have been generated. More recently, a tandem dimer of Keima (tdKeima) has been developed as the brightest version. Here we describe examples, which show the usefulness of Keima for dual-color fluorescence imaging technologies, such as fluorescence cross-correlation spectroscopy (FCCS) and two-photon laser scanning microscopy (TPLSM). Keima can be used in conjunction with existing fluorescent proteins in which the Stokes shift is much smaller, with the idea that while two fluorescent proteins are excited by a single laser each will fluoresce a different color.

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1. Introduction

Green fluorescent protein (GFP) was originally isolated from the light-emitting organ of the jellyfish *Aequorea victoria* by Shimomura et al. in 1962, although more than 30 years passed before the cDNA encoding the protein was subsequently characterized [1]. As *Aequorea* GFP is spontaneously fluorescent, chimeric GFP fusions offer the great advantage that they can be expressed in situ by gene transfer into cells, circumventing the need for high-level heterologous production, purification, *in vitro* labelling and micro-injection of recombinant proteins. In addition, these GFP fusions can be localized to particular sites within the cell by appropriate targeting signals.

Although spectral variants with blue, cyan, and yellowish-green emissions have been successfully generated from the *Aequorea* GFP [1], very few exhibit emission maxima longer than 529 nm [2]. Fortunately, the discovery of novel 'GFP-like' proteins mostly from *Anthozoa* (coral animals) have significantly expanded the range of colors available for cell biological applications [3]. Despite only a modest degree of sequence similarity, these GFP-like proteins share a β -can fold structure that is central to fluorescence in GFP.

Fluorescent compounds absorb light at the specific wavelengths, which provide enough energy to boost electrons to higher energy levels. Light is emitted as the electrons fall back to their normal (lower) energy levels. But some energy is lost as heat in the process, and so the wavelength of the light emitted is longer

(less energetic) than that of the light absorbed. The difference is known as the Stokes shift. We have developed a new GFP-like protein, which has an absorption energy typical of fluorescent proteins already in use, but a Stokes shift larger than usual, so that its fluorescence is easy to distinguish [4]. The protein was named Keima after a Japanese chess piece that moves in the hopping manner of a knight.

2. Description of methods

2.1. Protein fusion applications

Keima was developed based on a mutated non-fluorescent protein taken from the stony coral species, *Montipora* [4]. The original Keima forms a homotetrameric complex (referred to as 'tKeima'). At pH 7.4, the molar extinction coefficient (ϵ) at 440 nm and fluorescence quantum yield (Φ) of tKeima are $14,500 \text{ M}^{-1} \text{ cm}^{-1}$ and 0.22, respectively. Whether a fluorescent protein tends to form oligomers is an important consideration, as such interactions may interfere with the function of the host protein to which it is fused. First, the dimeric Keima (dKeima) was generated by introducing two mutations: V123T and V191I. Compared to tKeima, dKeima shows the same absorption spectrum peaking at 440 nm ($\epsilon = 24,600 \text{ M}^{-1} \text{ cm}^{-1}$), the same excitation and emission spectra with a Φ of 0.31. Then, after introduction of seven additional mutations (L60Q, F61L, V79F, T92S, T123E, Y188R, and Y190E), we successfully produced the monomeric Keima (mKeima), which exhibited the same spectra as tKeima and dKeima except that the emission maximum was 620 nm. The ϵ at 440 nm and the Φ of mKeima are $14,400 \text{ M}^{-1} \text{ cm}^{-1}$ and 0.24, respectively. mKeima is tolerant of fusion at the N- and C-termini.

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An alternative approach to protein fusions is to make a tandem dimer (td) of Keima. We have recently developed tdKeima by concatenating dKeima via a long linker. While tdKeima shows the same spectra with dKeima, it appears about three times brighter per molecules than mKeima; the value of count per molecule in fluorescence correlation spectroscopy for tdKeima was 2.7 times larger than that for mKeima.

To demonstrate the utility of tdKeima, we generated a mammalian expression plasmid that encodes a chimeric protein comprising β -tubulin, a linker, and tdKeima (Fig. 1a). Generally, the linker between a fluorescent protein and a host protein should be flexible, soluble, and resistant to proteolysis, and should not have a secondary structure or aggregate [5]. The amino acid that confers the greatest flexibility to a peptide chain is glycine, which has the smallest side chain. Thus, fusion proteins are often designed to include a number of glycine residues between the two components. The most widely used linker structures have sequences consisting primarily of stretches of Gly and Ser residues; the Ser residues are interspersed along the linker to improve the solubility of the poly-Gly peptides. Although sequences coding for these linkers are usually appended to PCR primers, it is laborious to prepare a long primer that includes Gly- and Ser-encoding stretches for each fusion protein. We have constructed a series of DNA plasmids that facilitate the fusion of coding sequences for any two protein domains with a DNA linker that encodes a triple repeat of the amino-acid linker Gly-Gly-Gly-Gly-Ser [(GGGGS)₃] [6]. The sequence encoding the linker was inserted into the EcoRV

site in the center of the multicloning site (MCS) of pBlueScript (pBS) (Stratagene). To account for all three potential reading frames and both orientations, six DNA plasmids were constructed and named pBS Couplers 1–6. Two sequential ligation reactions with a pBS Coupler using appropriate restriction enzymes allow the coding sequence of a fluorescent protein variant to be fused to the long linker and the protein of interest. We found that β -tubulin-(GGGGS)₃-tdKeima was incorporated into microtubules of HeLa cells (Fig. 1b).

2.2. Dual-color fluorescence cross-correlation spectroscopy

Dual-color fluorescence cross-correlation spectroscopy (FCCS) has several advantages over standard fluorescence correlation spectroscopy (FCS). While FCS detects molecular concentrations and mobility [7], FCCS enables the tracing of two spectrally distinguishable fluorophores, thus extracting essential information about the kinetics of molecular interactions [8]. FCCS employs two fluorophores of different colors. For their simultaneous excitation, conventionally two lasers are aligned to the same confocal spot. However, researchers often suffer from the difficulty in bringing two laser beams to a perfect and stable overlap. While FCCS can be performed using a single laser line (SL-FCCS) for single-photon excitation [9], this approach using common fluorophores that show broad excitation and emission spectra with modest Stokes shifts requires many mathematical considerations to compensate for cross-excitation, cross-talk, and fluorescence resonance energy transfer (FRET). Thus, simple SL-FCCS has been practicable only by using special fluorophores, including quantum dots and 'MegaSto-

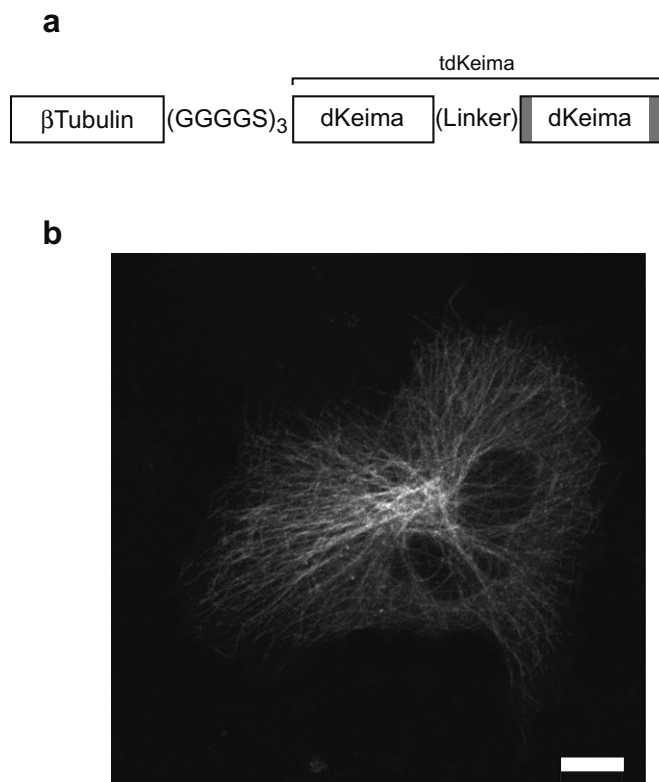


Fig. 1. Performance of tdKeima for protein fusion applications. (a) A schematic representation of the primary structure of β -tubulin-tdKeima. tdKeima is composed of two dKeima, which are fused via a linker. The shaded regions are coded by sequences with the humanized codon usage. Thus, the sequence for the whole tdKeima can be amplified by PCR. (b) Expression of β -tubulin-tdKeima in a HeLa cell. Scale bar, 10 μ m. Cells were imaged using the LSM 510 META system (Carl Zeiss) equipped an Ar laser. The excitation beam (458 nm) was reflected by a HFT458 dichroic mirror and focused by a water immersion objective lens (C-Apochromat 40 $\times$ /NA1.2; Carl Zeiss). The emission filter for mKeima was LP560.

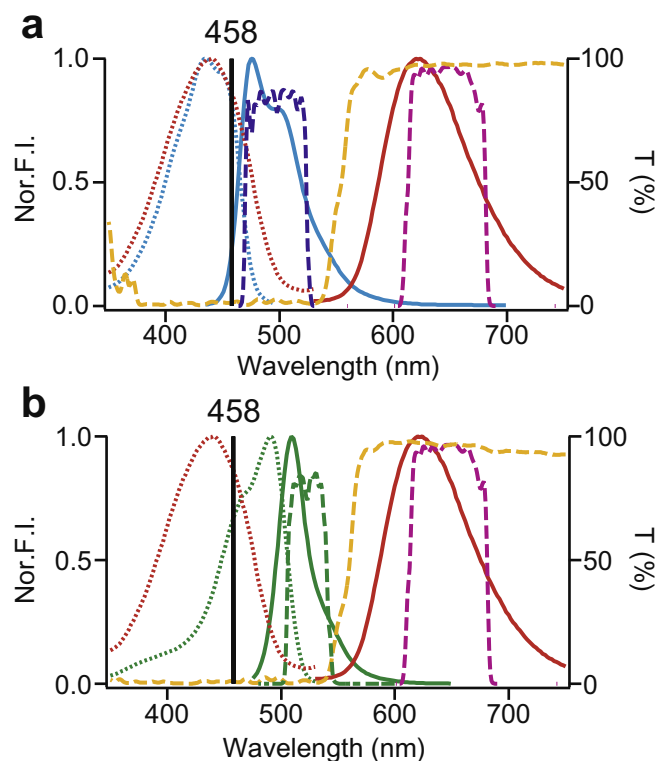


Fig. 2. (a) Fluorescence excitation (dotted line) and emission (solid line) spectra of mKeima (red) and ECFP (cyan). The wavelength of the laser line, 458 nm (Ar laser) is indicated by a vertical line. The wavelengths that pass through the emission filters and a dichroic mirror are shown (broken lines). (b) Fluorescence excitation (dotted line) and emission (solid line) spectra of mKeima (red) and EGFP (green). The wavelength of the laser line, 458 nm (Ar laser) is indicated by a vertical line. The wavelengths that pass through the emission filters and a dichroic mirror are shown (broken lines).

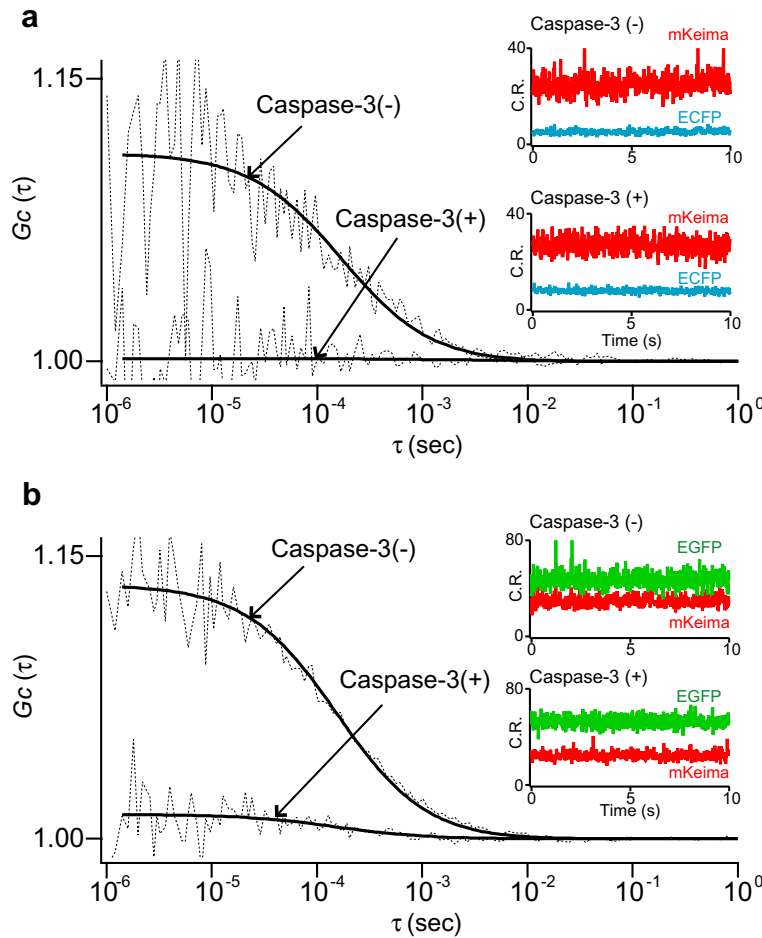


Fig. 3. Single laser wavelength (458 nm) excitation FCCS to monitor proteolysis by caspase-3 *in vitro*. (a) Cross-correlation analysis using mKeima and ECFP. Cross-correlation curves measured at 0 s (Caspase-3(-)), and 420 s (Caspase-3(+)) after the addition of caspase-3. (b) Cross-correlation analysis using mKeima and EGFP. Cross-correlation curves measured at 0 s (Caspase-3(-)), and 420 s (Caspase-3(+)) after the addition of caspase-3. $G_c(\tau)$: the cross-correlation function. Insets: the fluorescence intensities of mKeima and CFP in the two respective channels during an FCCS measurement. C.R., count rates.

kes' dyes (<http://www.dyomics.com>). These problems are serious for imaging technologies using fluorescent proteins because of the lack of a pair of fluorescent proteins with comparable excitation maxima but largely different Stokes shifts toward emission.

Combined use of mKeima with a cyan-emitting fluorescent protein (ECFP) allows for simple but efficient FCCS using a single laser line [4]. Due to complete separation of their emissions, this FCCS approach enabled sensitive detection of proteolysis by caspase-3. The carboxyl terminus of mKeima and the amino terminus of CFP (ECFP) were linked using a peptide containing the caspase-3 cleavage sequence DEVD. The recombinant protein (mKeima-DEVD-CFP) (10 nM) was incubated at 25 °C with activated caspase-3 (MBL) (0.2 U/ μ l) in buffer containing 20 mM Hepes-KOH (pH 7.5), 10 mM KCl, 1.5 mM $MgCl_2$, 1 mM EDTA, 1 mM EGTA, and 1 mM dithiothreitol. FCCS measurements were made using ConfoCor3 LSM510 META (Carl Zeiss) equipped with an Ar laser. The excitation beam (458 nm) was reflected by a HFT458 dichroic mirror and focused by a water immersion objective lens (C-Apochromat 40 $\times$ /NA1.2; Carl Zeiss). The emitted light was collimated and then split by a NFT545 dichroic mirror. Emission signals were detected through a BP615-680IR emission filter for ECFP and a BP470-495IR emission filter for mKeima. The acquired $G(\tau)$ function was fitted by a one-component model as

$$G(\tau) = \frac{1}{N} \left(1 + \frac{\tau}{\tau_1}\right)^{-1} \left(1 + \frac{\tau}{s^2 \tau_1}\right)^{-1/2}$$

where τ_1 is the diffusion time of the fluorescent particles, N is the average number of fluorescent particles in the excitation-detection volume defined by the radius w_0 and length $2z_0$, and s is the structure parameter representing the ratio $s = z_0/w_0$. The excitation and emission spectra of mKeima and ECFP with the passbands of the emission filters in Fig. 3a. Also, the transmittance of the dichroic mirror (NFT545) is superimposed. We strongly recommend researchers to make graphs, in which spectra are plotted as percent transmittance for the interference filters (excitation filters, emission

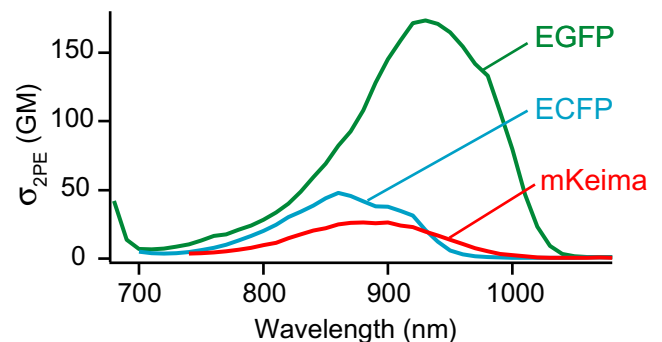


Fig. 4. Two-photon action cross sections (σ_{2PE}) of mKeima (red), ECFP (cyan), and EGFP (green).

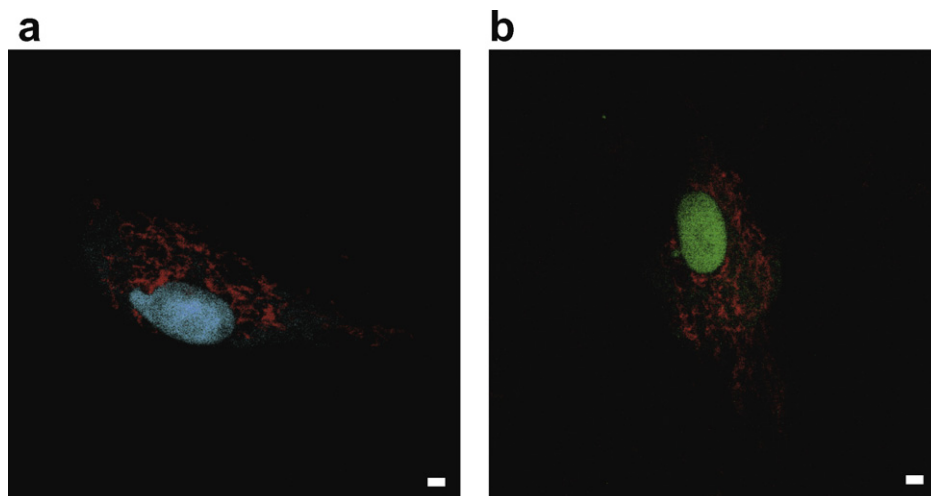


Fig. 5. (a) Visualization of the mitochondria and nucleus by dual-color TPLSM using mKeima and ECFP. The excitation wavelength was 880 nm. (b) Visualization of the mitochondria and nucleus by dual-color TPLSM using mKeima and EGFP. The excitation wavelength was 960 nm. Scale bar, 10 μ m.

filters, and dichroic mirrors) that are actually used, together with excitation and emission spectra of fluorescent dyes. The transmittance curves for the filters (normal incidence) and dichroic mirrors (45° incidence) can easily be measured using a conventional spectrophotometer. This measurement also provides the chance for researchers to check quality of interference filters, which may deteriorate with time.

Significant cross-correlation was observed between the fluctuations in the two detection channels (Fig. 2a, Caspase-3(–)). Incubation of the same sample with activated recombinant caspase-3 (0.2 U/ μ l) at 25 °C for 7 min almost completely abolished the cross-correlation signal (Fig. 2a, Caspase-3(+)).

Another appropriate partner of mKeima is EGFP, the most popular variant of *Aequorea* GFP. mKeima-DEVD-EGFP was generated and subjected to the cleavage experiment (Fig. 2b). Replacement of ECFP with EGFP changes slightly the optimal optical components for FCCS as illustrated in Fig. 3b.

2.3. Dual-color two-photon laser scanning microscopy

Two-photon laser scanning microscopy (TPLSM) offers important advantages over laser scanning confocal microscopy (LSCM), such as deep tissue penetration and three-dimensional localized excitation [10,11]. Since many spectral variants of FPs have emerged, more and more investigators are interested in dual-color imaging with TPLSM. We previously reported that although EGFP and mRFP1 (the monomeric version of DsRed) are widely employed for dual-color imaging experiments using LSCM, their σ_{2PE} values do not overlap very much [12]. Also, rapid exchange of the wavelength of an ultra-short pulse laser between the two respective optimal wavelengths is not practical. Therefore, it is difficult to determine the optimal wavelength for two-photon excitation of both EGFP and mRFP1.

We examined the combination of mKeima with ECFP or EGFP, since the single-photon excitation spectra of mKeima and ECFP

or EGFP match very well and their emissions are separable. However, the selection rules for two-photon processes are different from those for single-photon processes. We measured σ_{2PE} of mKeima and compared it with that of ECFP and EGFP. There were substantial spectral overlaps between mKeima and ECFP and between mKeima and EGFP [12] (Fig. 4).

mKeima-mit was constructed by fusing the 29 N-terminal amino acids of the cytochrome *c* oxidase subunit VIII presequence to the N-terminus of mKeima [12]. EGFP-nu and ECFP-nu were generated by extending EGFP and ECFP at the N-termini with the nuclear localization signal. cDNAs coding for the chimeric proteins were transfected into Vero cells using Lipofectamine 2000 (Invitrogen). Two-photon dual-color fluorescence imaging was performed using an upright microscope based on a scanning unit Fluoview FV1000 (Olympus; Tokyo, Japan). The objective lens used was LUMFL 60x/1.10w. ECFP and mKeima were excited at 880 nm, and their signals were detected using BA460-510 (Olympus), DM505 (Olympus), and 618/50 (Semrock) (Fig. 5a). EGFP and mKeima were excited at 960 nm, and their signals were detected using BA495-540HQ (Chroma), DM575 (Olympus), and 618/50 (Semrock) (Fig. 5b).

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